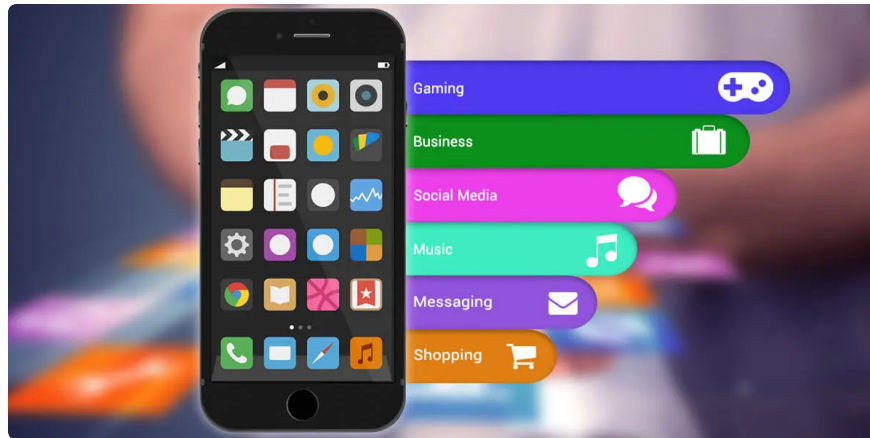


Year 2 Mini-Project

Image-Based English Vocabulary Learning Mobile App (Year 2)

2025–2026 (Proposed by Evelyn)



Project Description

Background

According to industry forecasts, the mobile app development market is projected to reach \$788.5 billion USD by 2030, driven by the explosive growth of smartphone usage and mobile-first digital experiences. This signals a paradigm shift: **the mobile-first era is not coming—it's already here.**

This project offers an excellent opportunity for students to deeply engage with key skills and mindsets in mobile development.

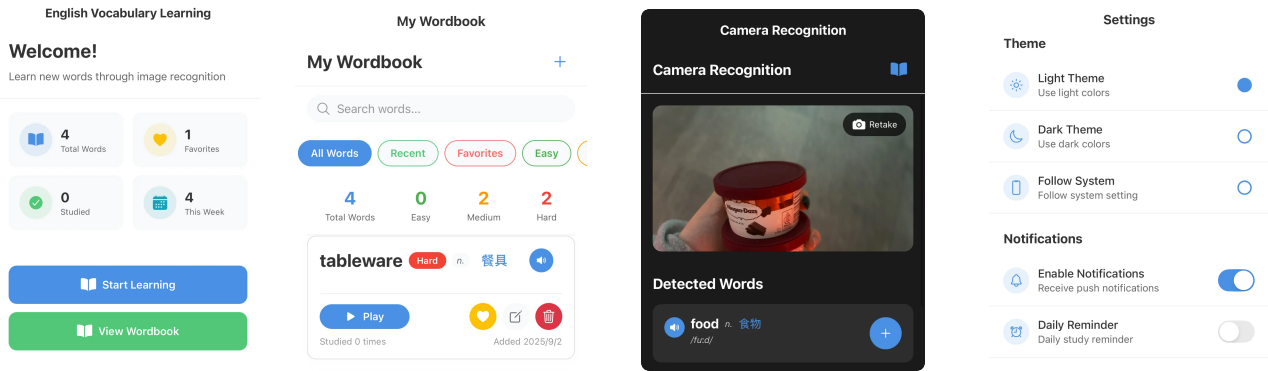
- Understand the key characteristics of mobile development
- Compare mobile development with web development
- Explore the unique advantages of mobile platforms

Project brief

Students should develop a mobile application for English learners

Image-Based English Vocabulary Learning Mobile App

This is a mobile app combining **image recognition** technology with **personalized vocabulary building**. Users can **take photos or upload images** to identify objects in their surroundings. The app then **provides the English word, phonetic transcription, and Chinese definition** for each recognized item. Learners can **save** these words to a **personal wordbook**, creating a **customized vocabulary list** based on real–world encounters



Main Tasks – 70%

- 1. **Camera Integration:** Integrate the device camera and enable image capture functionality
- 2. **Image Recognition:** Invoke an image recognition API to identify objects in the captured image
- 3. **Translation Service:** Translate the recognized English words into Chinese using a translation API
- 4. **Wordbook Management:** Implement a personal wordbook feature to store and categorize vocabulary
- 5. **Pronunciation Playback:** Enable pronunciation playback for each saved word
- 6. **Custom Profile:** Allow profile customization and theme switching

Bonus points – 20% (Details in the criteria page)

1. Translation service

Language Feature	Description
Pronunciation	Accurate phonetic transcription or audio link
Example Sentence	Example sentence(s) demonstrating usage
Part of Speech	Clear part–of–speech labeling (e.g., noun, verb)

2. README documentation

3. Personalized Function

You may choose to add any feature you wish, as long as it is **relevant to the app**. You can implement the following features (including but not limited to)

- Chatbot assistance
 - User login
 - Dashboard – Dynamic display of learning process
 -
-

Framework options

Not only that, you can choose according to your own preferences.

Native Frameworks

- **Swift** (for iOS)
- **Kotlin or Java** (for Android)

Cross-Platform Frameworks

- **Flutter** (Dart)
 - **React Native** (JavaScript)
 - ...
-

NOTE !

Students may choose to develop the app using either **native development** or a **cross-platform** framework. If opting for native development, supporting a single platform (e.g., Android or iOS) will receive the **same score** as implementing both platforms via a cross-platform solution. Deployment is **not required**—local execution and demonstration of core functionality is sufficient.

Useful Materials:

The tutorials and reference links provided below are **examples, not universal solutions**. Their usefulness depends on the **framework you choose**:

Choose which framework?

- React Native vs Flutter In 2025 – https://youtu.be/M_z6shjaoNw?si=IfRrv110s9DrS5n6
- React Native vs Flutter – Which should you use? – https://youtu.be/AiG8-wHidiA?si=wVCBT-092_wLtgNy
- Native/Cross-platform – <https://circleci.com/blog/native-vs-cross-platform-mobile-dev/>

Tutorial

- Learn what frontend, backend and API means in simple words – <https://youtu.be/W1T0rLGz9Jw?si=jWTJ9RF59pU3SePi>
- APIs for Beginners – How to use an API – <https://youtu.be/WXsD0ZgxjRw?si=cnJq7vRDnRDnSTS>
- Build a React Native App – <https://youtu.be/0-S5a0eXPoc?si=0JFRS6G8pmRywjOR>
- React Native Full Course for Beginners – <https://youtu.be/WDunoPNBxKA?si=PSd1NvxsVXOBQeHH>

AI-assisted code development

- Build Your Mobile App In 22 Minutes – <https://youtu.be/ADgIkx5NHJ8?si=Pj7roA3LsxcJstwT>
- Cursor AI Tutorial for Beginners1 – <https://youtu.be/3289vhOUdKA?si=HWnspiuhQWfiqp82>
- Cursor AI tutorial for beginners2 – <https://youtu.be/gqUQbjsYZLQ?si=1clqAm7ZRHinAwXr>

API reference

- Image Recognition
 - Google Cloud Vision API – https://cloud.google.com/vision?hl=zh_cn
 - Microsoft Azure Computer Vision – <https://learn.microsoft.com/en-us/azure/ai-services/computer-vision/how-to/call-analyze-image?tabs=rest>
 - Amazon Rekognition – <https://docs.aws.amazon.com/rekognition/latest/dg/what-is.html>
- Translation service
 - Google Cloud Translation API – <https://cloud.google.com/translate/docs/reference/rest/>
 - Youdao Translation API – <https://ai.youdao.com/product-fanyi-text.s>

Generative Artificial Intelligence Usage Policy

Acceptable Use of GenAI

Students may use GenAI tools for the following purposes, provided they are *transparent and reflective* about their use:

- Idea generation (e.g., brainstorming project topics or design approaches)
- Code assistance (e.g., debugging, syntax suggestions)
- Language support (e.g., grammar correction, paraphrasing)
- Visualization (e.g., generating concept diagrams or mockups)

Prohibited Use

The following uses of GenAI are not permitted:

- Submitting AI-generated content as original work without disclosure
- Using GenAI to write substantial portions of the report, literature review, or analysis
- Generating falsified data or results
- Using AI to bypass assessments or simulate experimental work

Disclosure Requirements (FILL THE FORM)

Students must include an AI Usage Statement in their work in any form, detailing:

- Tools used (e.g., ChatGPT, Copilot, etc)
 - Purpose and context of use
 - Reflections on accuracy, limitations, and ethical considerations
-

Final Code Submission Guidelines

1. Required Submission Items

- a. Source Code – Mandatory
- b. FCA Declaration on GenAI – Form
- c. README File – if you have

2. Submission methods

- a. Upload to GitHub – Then share the link with me. (Via WeChat/Email/...)
 - b. Send Zip via Email – Email the file directly to p2321167@mpu.edu.mo
-

Reminders for Your Project Journey

1. Attendance & Meetings

You are expected to attend scheduled project meetings. During each meeting, you can share your current progress and discuss any challenges you're facing.

- Your performance on the meetings will **not be graded**, but they reflect your **attitude, engagement, and communication skills**.
- We also need to prepare reports based on your attitude, so please take them seriously.

2. AI Is a Tool — Not a Substitute for Your Thinking

You are allowed to use AI tools to assist with coding or design. However, AI cannot replace your understanding.

Even if your code is AI-assisted, you must **understand how it works** and be able to **explain** it.

In the **final interview(Oral)**, you will be asked about **how your features were implemented** — so make sure you know your own project.

3. You Can Choose Your Challenge

If this project feels overwhelming, you may choose to work on the Year 3 project (Web + AI Chatbot). – **Because** Lower grade students are allowed to undertake the projects of higher grades.

This project might be the most difficult of the three. But if you're ready to take on a challenge, with AI support and your own persistence, you'll gain valuable skills and confidence.

Final Words

I wish each of you success — not just in completing the project, but in discovering what you're capable of.

Marking Criteria of Mini Project (Year 2)

2025–2026 (Evelyn)

Marking Criteria (Total 100%)

The marking criteria is based on the **attendance of meetings** and **final submitted files**.

Students should accomplish all the necessary tasks and submit the files **before deadline**.

1. Attendance of Meetings – 10%

- If a student attends **all the meetings**, he or she will gain **10%**.
- **2%** mark will be **deducted** for **every missing meeting**.
 - If a student is unable to attend a scheduled project meeting due to special circumstances, they must explain the reason in advance and provide progress updates through alternative means.
 - Recording a short video update on project progress
 - Communicating online (e.g., Zoom, Teams, or WeChat)
- If someone finishes earlier, he or she will also gain 10%. If someone **misses all the meetings**, he or she will gain **0%**.

2. Project – (70%+20% Bonus)

The project is evaluated in two parts – **Core Features** and **Bonus Features**.

- ◆ **Core Features** – Total possible score: **70 points**
- ◆ **Bonus Features** – **Maximum** bonus score: **20 points**

UI & Navigation Layer – 15%

1. Navigation System – 7%



This is just an example. Just do it according to your own ideas.

Evaluation Item	Points	Description
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No navigation implemented	0 %	No page-to-page navigation
Basic navigation structure implemented	5 %	Allow users to move between multiple pages.
Clear navigation logic and user-friendly UI	7 %	Clear element arrangement, mobile-friendly

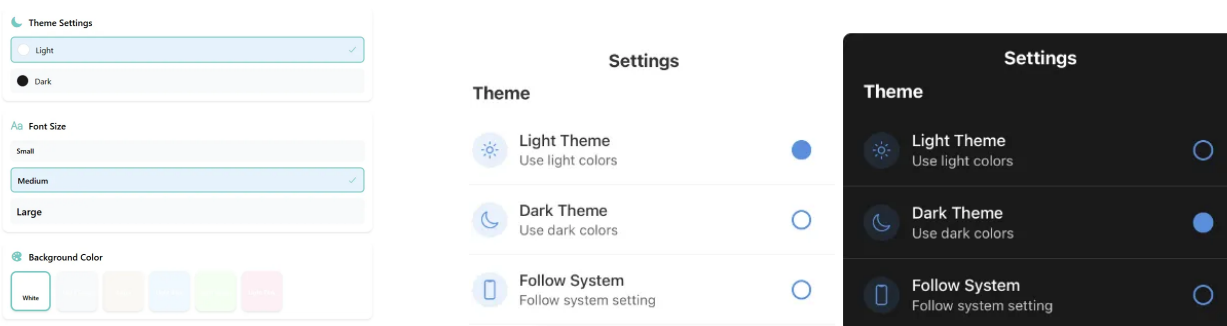
- *Sample for good navigation* – <https://youtu.be/r8m2wM7eXeU?si=qxdgGyRFUv23BkLe>
- *Navigation (Android)* – <https://youtu.be/mlohTkl7MaI?si=BC6oNBKGr0by9PCI>
- *Tab Navigation (React Native)* – <https://youtu.be/zEbaeod2QjM?si=K6BWoeDsQXzpTwLm>

2. User customization feature – 8 %

Students may freely choose the type of user customization features to implement, **including but not limited to** the following:

- Theme switching (light/dark mode)
- Custom background image or color
- Avatar upload and replacement
- Language preference settings (e.g., English/Chinese toggle)
- Font size or style adjustment

User customization features may be implemented in different forms – **No restrictions** on the implementation method :



- One-click button on the home screen
- Standalone settings page
- ...

Evaluation Item	Points	Description
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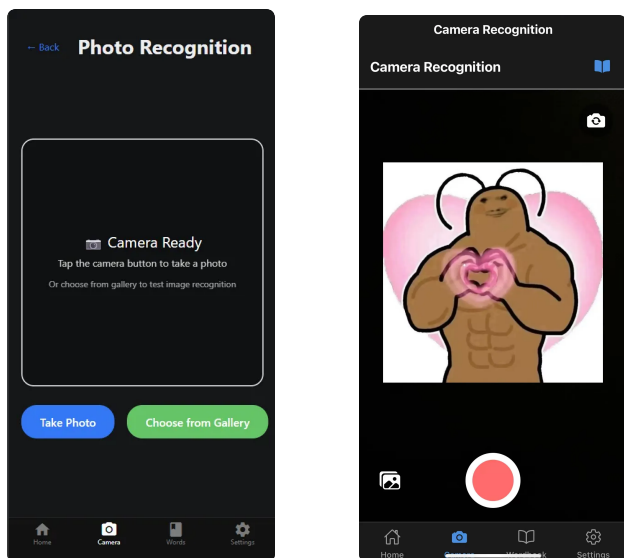
Page supports user customization (e.g., theme, avatar, background, language preferences)	10%	Fully functional, reflects user personalization, responsive UI
Page missing or inaccessible	0%	Cannot be evaluated, considered incomplete

NOTE !

- Students only need to implement one personalized customization function!
- Multiple functions **will not earn** additional points.

Client-side Logic Layer – 55%

1. Camera Integration – 10%



Description	Points
No camera access or capture functionality	0%
Camera opens but no image capture or processing	5%
Fully functional; image captured and usable for next steps	10%

- Flutter Camera App Tutorial : <https://youtu.be/TrmoRtn5MZA?si=CkDZYsAOq7YDqBNl>
- Android Camera Integration: <https://youtu.be/DPHkhamDoyc?si=cYVqj2wRJltpWSEh>
- Expo Camera in react native expo : <https://youtu.be/zvYc5Ar8ni8?si=BaRv00hrj34DA4os>

2. Image Recognition – 10%

Description	Points
No recognition API or response	0%
Accurate recognition with clear result display	10%

API reference:

- Google Cloud Vision API – https://cloud.google.com/vision?hl=zh_cn
- Microsoft Azure Computer Vision – <https://learn.microsoft.com/en-us/azure/ai-services/computer-vision/how-to/call-analyze-image?tabs=rest>
- Amazon Rekognition – <https://docs.aws.amazon.com/rekognition/latest/dg/what-is.html>

(Not limited to these, you can choose the API that suits you best)

3. Translation Service – (10+10)%

Description	Points
No translation API or result	0%
Accurate translation with visible UI output	10%

Students will earn **10%** if they use a translation API to correctly display **at least one English meaning along with its accurate Chinese translation**.

The requirement is fulfilled by showing just one valid word pair (e.g., “apple — 苹果”), without needing multiple entries or extended functionality.

- Tutorial

React Native Tutorial for Beginners – Build a Dictionary App – https://youtu.be/A1_j0BR3dz8?si=Q_Rwef0wUf_wyVpq

API reference:

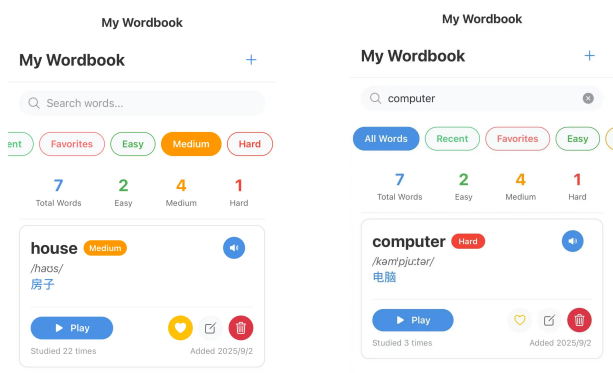
- Google Cloud Translation API – <https://cloud.google.com/translate/docs/reference/rest/>
- Youdao Translation API – <https://ai.youdao.com/product-fanyi-text.s>

(Not limited to these, you can choose the API that suits you best)

Bonus points

Language Feature	Description
Pronunciation	Accurate phonetic transcription or audio link
Example Sentence	Example sentence(s) demonstrating usage
Part of Speech	Clear part-of-speech labeling (e.g., noun, verb)
Bonus Rule	One feature: 5% Two/ALL features: 10%

4. Wordbook Management – 20%



Content		Score
Implement a personal wordbook feature to store and categorize vocabulary		Up to 20%
Visually appealing layout with clear structure and user-friendly interaction		2%
Each vocabulary item displays both English meaning and accurate Chinese translation		5%
CRUD	Users can add vocabulary entries	One feature implemented –5% Both features implemented – 8%
	Add refers to selecting a recognized word and saving it to the personal wordbook.	
	Users can delete vocabulary entries	
Supports categorization by tags or difficulty level		One feature implemented –3%

Implement the word search function	Both features implemented – 5%
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5. Pronunciation Playback – 5%

Description		Score
Playback works	Pronunciation can be played by clicking a button or icon	3%
Icon is clearly visible and intuitive	Playback is triggered via a recognizable icon (e.g., speaker symbol)	5%
Neither works nor visible	No playback or unclear trigger	0%

Note: If both playback and icon are implemented correctly, the full 5 points are awarded. If only playback works but the icon is missing or unclear, award 3 points. If neither is functional, award 0.

- *Playback for Expo React Native Apps* :https://youtu.be/pd_Ez9Kbi2c?si=qKSpd0Q5mjEK3ix8
- *Flutter Audio Pronunciation* :https://youtu.be/kUeTVggUJTtw?si=UX1kqChtdmpH3__N

6. Personalized Function **【Bonus】**

You may choose to add any feature you wish, as long as it is **relevant to the app**. You can implement the following features (including but not limited to)

- Chatbot assistance
- User login
- Dashboard – Dynamic display of learning progress
-
- **Each completed feature** can earn up to **10%**

Criteria	Description	Score Weight
Feature Completion	The feature is fully implemented and functional within the app	8%
Textual Explanation of Function	Clear and appropriate explanation of what the feature does and how it works	10%

7. README Documentation – 5% **【Bonus】**

The **README.md** file is a project overview and guide and you need to use **Markdown syntax** for clear structure.

It may include

- Project Overview – Brief summary of your app
 - Framework Declaration – Which framework do you use?
 - Setup Instructions
 - Feature Summary – List of core and bonus features implemented, with short descriptions
 - ...
-

Bonus Scoring Rules

No matter how many bonus features are completed, the maximum bonus score is capped at 20%

Project Evaluation Table

Feature		Criteria / Description	Max Points
UI & Navigation Layer (15%)	Navigation System	No navigation: 0%	7
		Basic structure: 5%	
		Clear logic & mobile-friendly UI: 7%	
	User Customization Feature	No functional customization: 0%	8
		One functional customization: 8%	
Subtotal			15
	Camera Integration	No access: 0%	10
		Opens but no capture: 5%	
		Fully functional: 10%	

Client-side Logic Layer (55%)	Image Recognition	No API or result: 0%	10
		Accurate recognition & clear display: 10%	
	Translation Service	No API or result: 0%	10
		Correct English–Chinese pair shown: 10%	
	Wordbook Management		20
	Pronunciation Playback	Playback only: 3%	5
	Playback + clear icon: 5%		
Subtotal			55
Bonus Features (Max 20%)	Language Feature Enhancements	One feature: 5%	10
		Two or all: 10%	
	Personalized Function	Each feature with a textual explanation: 10%	20
	README Documentation		5
Subtotal (MAX)			20
Total Project Score			90